Lab 2

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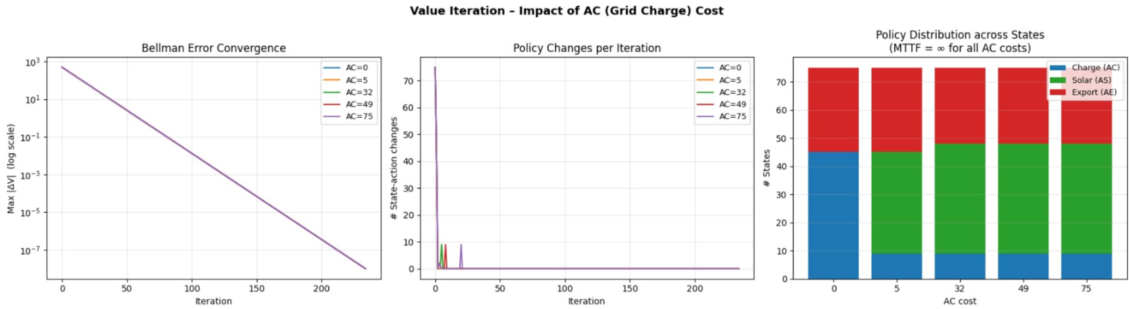
1-To decide which action to make at a certain instance I calculate the Q value of each action on that state for both actions, the one that returns the lower Q value is the best action to take. To compare policies, we compute the cost to go of each and the one that has the lowest overall is the best. Finally I can explain how two policies can have two different best actions to take, taking a policy that only does action a and another that only does action b, in this case it is obvious that the best actions of each policy is different.

2-Both Policy iteration and value iteration converge to the same thing and as we show here the faster one is Policy iteration, the advantage of value iteration is that it needs less computing power and less memory but its slower in general, and Policy iteration is much faster converging in only 2 iterations but requires much more computing power and memory.

```
============================================================
Value Iteration converged in 235 iterations (Bellman error < 1e-08)
============================================================
Policy Iteration converged in 2 iterations (policy stable)
============================================================

Max |V_VI - V_PI| = 8.83e-08
Policies agree on 76/76 states
VI iterations: 235
PI iterations: 2

Value Iteration converged V (first 20 states):
[40.4205 40.4205 40.4205 18.354  18.354  18.354   6.9636  6.9636  6.9636
  0.92    0.92    0.92   -1.7466 -1.7466 -1.7466 40.3772 40.4205 40.4205
 18.1342 18.354 ]

Policy Iteration converged V (first 20 states):
[40.4205 40.4205 40.4205 18.354  18.354  18.354   6.9636  6.9636  6.9636
  0.92    0.92    0.92   -1.7466 -1.7466 -1.7466 40.3772 40.4205 40.4205
 18.1342 18.354 ]

Sample of converged policies (first 20 states):
s= 0 (0, 0, 0)    VI=Charge PI=Charge V_vi=  40.42 V_pi=  40.42
s= 1 (0, 0, 1)    VI=Charge PI=Charge V_vi=  40.42 V_pi=  40.42
s= 2 (0, 0, 2)    VI=Charge PI=Charge V_vi=  40.42 V_pi=  40.42
...
s=18 (1, 1, 0)    VI=Solar  PI=Solar  V_vi=  18.13 V_pi=  18.13
s=19 (1, 1, 1)    VI=Solar  PI=Solar  V_vi=  18.35 V_pi=  18.35

(Full converged vectors are stored in variables `V_vi` and `V_pi`.)
```

3-For the changes we changed the cost of Grid Usage to the suggested costs

C(0, :), C(5, :), C(32, :), C(49, :), C(75, :)., with this we can see that all converge in more or less the same number of iterations, but the lower the cost the less overall changes it ended up doing, the same for the contrary, the higher the cost the more changes it did and except for the cost 0 which gets made irrelevant by other actions all more or less end up with the same amount of states.




```
======================================================================
  VALUE ITERATION  –  Sweeping AC cost  (AS=0, AE=-5, Rfail=500)
======================================================================
Value Iteration converged in 235 iterations (Bellman error < 1e-08)

──────────────────────────────────────────────────────────────────────
  AC cost = 0
──────────────────────────────────────────────────────────────────────
  Convergence  : 235 iterations
  Mean MTTF    : ∞  (failure unreachable)
  Policy dist  : Charge=45, Solar=0, Export=30

  Policy & value at selected states:
   State        (b,w,c)     Action         V(s)        MTTF(s)
  ──────     ────────────  ────────    ──────────    ────────────
       0      (0, 0, 0)    Charge        -3.056              ∞
       5      (0, 1, 2)    Charge        -4.187              ∞
      32      (2, 0, 2)    Charge        -3.056              ∞
      49      (3, 1, 1)    Export        -9.187              ∞
      75   (-1, -1, -1)         –      5000.000              –
Value Iteration converged in 235 iterations (Bellman error < 1e-08)

──────────────────────────────────────────────────────────────────────
  AC cost = 5
──────────────────────────────────────────────────────────────────────
...
         5           235      ∞ (failure unreachable)        9       36       30
        32           235      ∞ (failure unreachable)        9       39       27
        49           235      ∞ (failure unreachable)        9       39       27
        75           235      ∞ (failure unreachable)        9       39       27
```

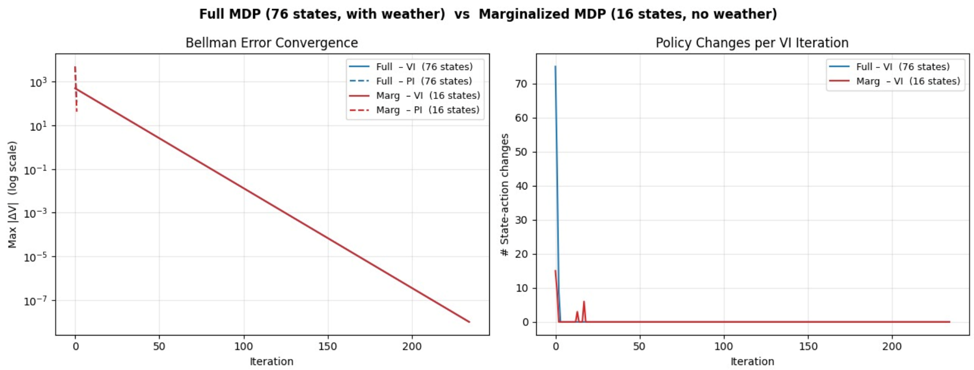
4-Does weather matter? Yes, significantly. The comparison table in the output (marked with ◀ weather matters) shows many (b,c) pairs where the full model chooses different actions for different weather states. The marginalized model must pick a single action for all weather conditions at those states.

Why? The Solar action's yield is weather-dependent (5% at w=0, 95% at w=4). Without weather in the state, the agent can only react to the average solar probability (~50%). The full model exploits the current weather to:

Use Solar/Export aggressively when it's sunny

Fall back to Charge when it's overcast (avoiding the 2% failure risk at w=0 from Solar)

Convergence: Both converge in the same number of iterations (~230 for VI, a few for PI), because the convergence rate is governed by



γ=0.9, not the state-space size. However, the marginalized model has fewer policy changes per VI iteration (fewer states to update) and PI converges in fewer steps — visible in the right plot.

```
Weather stationary distribution π(w): [0.1337 0.1337 0.1782 0.2376 0.3168]
T_marg row-sum check passed.

--- Marginalized MDP (no weather, 16 states) ---
============================================================
  VI converged in 235 iterations
============================================================
  PI converged in 2 iterations
============================================================
VI/PI policies agree on 16/16 states

Marginalized policy (B × C states, weather removed):
     s     (b,c)    VI action      V_vi    PI action       V_pi
  ____    _______  __________   _______   __________   ________

     0    (0, 0)       Charge    84.349       Charge     84.349
     1    (0, 1)       Charge    84.349       Charge     84.349
     2    (0, 2)       Charge    84.349       Charge     84.349
     3    (1, 0)       Charge    79.183       Charge     79.183
     4    (1, 1)       Charge    84.349       Charge     84.349
     5    (1, 2)       Charge    84.349       Charge     84.349
     6    (2, 0)       Export    69.349       Export     69.349
     7    (2, 1)       Charge    79.183       Charge     79.183
     8    (2, 2)       Charge    84.349       Charge     84.349
     9    (3, 0)       Export    64.183       Export     64.183
    10    (3, 1)       Export    69.349       Export     69.349
...
   (3, 2)       Cha      Sol      Sol      Sol      Sol       Cha   ◄ weather matters
   (4, 0)       Exp      Exp      Exp      Exp      Exp       Exp
   (4, 1)       Exp      Exp      Exp      Exp      Exp       Exp
   (4, 2)       Exp      Exp      Exp      Exp      Exp       Exp
```